

### Topic 35: Electricity – Resistance, current and voltage

| Students should: |                                                                                                     | Guidance<br><i>iLowerSecondary Curriculum reference</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 35.1             | know what is meant by electrical resistance and that it is measured in ohms ( $\Omega$ )            | <p>understand that components such as bulbs and resistors make it more difficult for a current to flow through have a high resistance; components such as copper wire that are easy for a current to flow through have a low resistance</p> <p>(electrical) resistance of a component indicates how hard it is for current to flow through a component</p> <p>understand that the higher the total resistance of the components in a circuit, the smaller the current that flows</p> <p><i>(This builds on Year 7: Electricity – Resistance)</i></p> <p><i>Year 9: Electricity – More on resistance, current and voltage</i></p> |
| 35.2             | know the relationship: voltage (V) = current (I) x resistance (R) and perform calculations using it | <i>Year 9: Electricity – More on resistance, current and voltage</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |